Interview Question Bank

Section-wise Compilation from Candidate Feedback

11 Sections

143+ Questions

What's Inside

Section 1

Self Introduction & HR

12 questions

Self Introduction & HR

Q1 Self introduction / Introduce yourself

Q2 Tell me about yourself

Q3 Explain about your projects / project details

Q4 About internship / certificates

Q5 Why did you choose Python / this language?

Q6 What are your hobbies?

Q7 Leadership qualities

Q8 Where do you see yourself in 5 years?

Self Introduction & HR

Q1 Why this company?

Q2 Any questions for the interviewer?

Q3 ID check / verification

Q4 Was money demanded during the process?

Section 2

Python – Core Concepts

16 questions

Python – Core Concepts

Q1 Difference between list and tuple

Q2 Difference between sets and dictionaries

Q3 Data types in Python

Q4 Lambda functions (with code example)

Q5 List comprehension with code example

Q6 Deep copy vs shallow copy

Q7 Exception handling (try, except, finally blocks)

Q8 About arguments: *args and **kwargs

Python – Core Concepts

Q1 Negative indexing in Python

Q2 What is Python and its features?

Q3 Is Python compiled or interpreted?

Q4 What is PEP? What is GBL (GIL)?

Q5 Built-in functions in Python

Q6 Generators and decorators

Q7 Python libraries (e.g., for plotting graphs)

Q8 Difference between / and //

Section 3

Python – Coding Questions

18 questions

Python – Coding Questions

Q1 Reverse a list / string (without built-in functions / without slicing)

Q2 Reverse a string with special characters in place

Q3 Reverse words in a string (e.g., 'hello world' → 'world hello')

Q4 Palindrome check (string and number, in 2 lines)

Q5 Fibonacci sequence (code)

Q6 Factorial (using recursion)

Q7 Swap two numbers without a third variable

Q8 Find min and max without built-in functions

Python – Coding Questions

Q1 Print second largest element from a list

Q2 Write a code for prime number validation

Q3 Find frequency of each character in a string

Q4 Remove duplicate elements from a list

Q5 Sort a list / find common elements in two lists

Q6 Write a lambda expression to print even elements

Q7 Number divisibility check (extract last digits)

Q8 Code for palindrome and even number combined check

Python – Coding Questions

Q1 Print largest of three numbers

Q2 Find missing number in an array sequence

Section 4

Object-Oriented Programming (OOP)

12 questions

Object-Oriented Programming (OOP)

Q1	OOP concepts (4 pillars: Encapsulation, Polymorphism, Abstraction, Inheritance)
Q2	Real-world examples for all four OOP pillars
Q3	What is inheritance? Show example in Python/Java
Q4	Can you implement inheritance without a subclass?
Q5	Types of polymorphism
Q6	Overloading vs overriding difference
Q7	Abstract class vs Interface
Q8	Can static / private methods be overridden?

Object-Oriented Programming (OOP)

Q1 Constructor vs methods

Q2 Encapsulation – default values of instance and local variables

Q3 Garbage collection in Java

Q4 Wrapper classes in Java

Section 5

Java – Concepts & Coding

23 questions

Java – Concepts & Coding

Q1	What is Java and its features? Rate yourself in Java
Q2	JDK, JRE, JVM – differences
Q3	Is Java compiled or interpreted?
Q4	JIT (Just-In-Time compiler)
Q5	Explain public static void main and method-related questions
Q6	Is main method without static possible?
Q7	final, finally, finalize – differences
Q8	throw vs throws; exception handling in Java

Java – Concepts & Coding

Q1	try-catch code implementation
Q2	Inheritance code in Java
Q3	Collections framework
Q4	LinkedList vs ArrayList difference
Q5	HashMap vs HashTable difference
Q6	Array vs ArrayList difference
Q7	== vs .equals() (with pseudo code output)
Q8	Packages in Java (difference between modules and packages)

Java – Concepts & Coding

Q1 Factorial in Java (and JavaScript)

Q2 Print largest of three numbers in Java

Q3 Prime number code in Java

Q4 Palindrome code in Java (with and without for loop)

Q5 Inverse of a binary tree

Q6 Perfect square code

Q7 Frequency of characters in a string

Section 6

SQL – Concepts

15 questions

SQL – Concepts

- **Q1** What is SQL and its uses?
- **Q2** DDL, DML, DQL, DCL, TCL commands
- **Q3** SQL constraints
- **Q4** Primary key vs foreign key
- **Q5** Keys in SQL (all types)
- **Q6** What is normalisation?
- **Q7** Indexing in SQL (with code)
- **Q8** What is a view in SQL?

SQL – Concepts

- **Q1** ACID properties
- **Q2** Aggregate functions in SQL
- **Q3** String types in SQL / difference between FLOAT and DOUBLE
- **Q4** DELETE vs TRUNCATE vs DROP
- **Q5** UNION vs UNION ALL
- **Q6** Exist vs IN
- **Q7** DISTINCT and self join

Section 7

SQL – Joins & Queries

12 questions

SQL – Joins & Queries

- **Q1** Types of joins and their descriptions
- **Q2** Inner join vs left outer join
- **Q3** Difference between join and union
- **Q4** Rank() vs dense_rank()
- **Q5** SQL query: second highest salary
- **Q6** SQL query: top 10 highest salaries
- **Q7** SQL query: top 10 rated restaurants
- **Q8** SQL query: fetch details from a table

SQL – Joins & Queries

Q1	Inserting a new column into an existing table (ALTER)
Q2	How to view duplicates in SQL
Q3	Create employee table with given parameters
Q4	Inner join query example

Section 8

Web Technologies

11 questions

Web Technologies

Q1	What is HTML? List 5 tags
Q2	HTML class, action, attributes, forms, form types
Q3	Difference between div and span
Q4	HTML lists
Q5	What is CSS and its types?
Q6	CSS selectors
Q7	Difference between display:none and visibility:hidden
Q8	JavaScript form validation

Web Technologies

Q1 How do you describe String in Java?

Q2 Difference between Java and JavaScript

Q3 What is web server? (web servers you worked on)

Section 9

Cloud Computing

5 questions

Cloud Computing

Q1 What is cloud computing?

Q2 Benefits of cloud computing

Q3 Public cloud vs private cloud

Q4 Cloud infrastructures (IaaS, PaaS, SaaS)

Q5 What database did you use in your projects?

Section 10

Aptitude & Situational

12 questions

Aptitude & Situational

Q1 Aptitude: number series and find missing numbers

Q2 Aptitude: probability problem

Q3 Aptitude: train problem

Q4 Aptitude: time and distance

Q5 Aptitude: solve expression and find ratio

Q6 Aptitude: ATM transaction code / transpose of a matrix

Q7 Situation: How do you automate a repeated daily task?

Q8 Situation: How do you handle pressure at work?

Aptitude & Situational

Q1 Situation: How do you solve conflicts between teammates?

Q2 Situation: How would you work on a project with less knowledge of a technology?

Q3 Situation: If you can complete a project before deadline, what would you do?

Q4 Situation: How would you convince friends to implement your ideas?

Section 11

Willingness & Preferences

7 questions

Willingness & Preferences

Q1 Are you willing to relocate?

Q2 Are you flexible for any shift (including night shifts)?

Q3 Are you comfortable learning new technologies?

Q4 Are you comfortable switching technologies?

Q5 Are you doing any current internship?

Q6 Do you have a preferred location?

Q7 Why did you choose Python/Cybersecurity/this cluster?